Humanoids Cost as Much as an SUV Now | Nikhil Kamath x Brett Adcock | WTF Online Ep 2

Let me know when the guys can start. Ready? Would you leave your kids with your humanoid? We're not there today. Like I would not let my robot roam free for hours and weeks right now with my young kids.

Everybody talks about China is so far ahead in robotics. Why is that? Yeah, I just don't believe this. Hi, Brett, lovely to meet you.

What's behind you? Maybe we can start there. Yeah, so I'm at figure offices. We design humanoid robots.

So these are robots that kind of like move around the world like humans, legs, arms, hands. Behind me is our first generation robot, second and third generation humanoid robots here at figure. OK, so tell us a bit about yourself, Brett.

Where did you begin? Where were you born? How did you come to build a humanoid? Yeah, so I was born in the Midwest here in the US on a third generation farm, actually. We farm corn and soybeans. And then basically since like middle school and high school, I started working on technology businesses and mostly in software.

And that led to basically now 20 years building tech companies. I spent about 10 years in software, ended up starting a company in 2012 called Vetteri, and it was an AI marketplace for recruiting. The goal is to basically help candidates find jobs and employers find great talent.

And that whole process is extremely broken and hard. I ended up selling Vetteri for a little over 100 million in 2017 to the Deco Group, which is like the largest recruiting company in the world. After that, I really want to get into hardware.

So I started a company called Archer Aviation in 2018. And what we do is we build electric aircraft. Electric vertical takeoff and landing aircraft.

So it's kind of like it takes off kind of like a helicopter, so you can put them in cities and it flies a little bit like an airplane and it's fully electric power train. And the goal is to basically help move traffic that's like, you know, in cities and move into the air so folks can get around a lot easier. And yeah, basically design five generations of aircraft.

So far at Archer, our current aircraft is Midnight, it's a piloted four passenger aircraft that we're flying now, trying to get that certified in the U.S. airspace with FAA. Ended up taking that company public about four years ago, five years ago. And then, yeah, three or three years ago, I started Figure.

So we design and manufacture humanoid robots. And the goal here at Figure is to be able to do

everything a human can in the physical world. I have an investment in a company here in India, which does electric taxis called Sarla Aviation.

How far are we really? Traffic is a big problem in India. And, you know, I don't know how different Archer is because I've not seen enough about it. But it's the typical electric taxi, 160 kilometers range on a charge, meant to take off from hotels, the helipads on top of a hotel, get to the airport.

How far are we from a world where this could be more pragmatic? This is something that will be used by people. I mean, we have aircraft today that are that are flying fully piloted. You have a couple of components here.

The first is you have to be in the right decade to make sure the technologies are all feasible to make this really happen. This is a really hard technology. Most of the problems and challenges here happen because we have a there's probably like a 30 X difference in energy density between like traditional fossil fuel, say kerosene, that you might find in an aircraft, in an electric aircraft like what we find in batteries.

There is the problem just like building that technology and really making it viable. I think we've shown at Archer that that is like fully possible today. I mean, we fly basically every week.

Our full scale aircraft at this point, fully electric, similar concept operations we'd want to do in market. At this point, it's largely like a safety and certification and policy. It's like basically like what's getting this is certification.

In most countries, it's governed federally where you need like, you know, in Europe, it's YASA. Here in the US, it's FAA. You need you need like basically a certification level at the federal level that unblocks your ability to take passengers and make it ubiquitous.

So at this point, like largely missing certification, you didn't have like manufacturing and scale needed. But the technology is certainly here. So I think within the next like five years, we'll be taking passengers like paying passengers around in airspaces here in the world on electric aircraft like this.

And if that's true for America, do you think that will be the way forward? Do you think that will replace human driven taxis in the near future? I don't think it replaces all human driven taxis. I think it just takes a large portion of what's like the basically the folks that are taking transportation today that might be going that might be spending like half an hour, hour, hour and a half in traffic, which is sizable in some places like L.A. It's like almost 10 percent of like all the all the travel in the mornings and after work. It's just a much better quality of life if we're not sitting in traffic all day.

Would it also change short haul flights like say between San Francisco and L.A., for example, which is one hour? Like my view is like all aircraft, like all aviation, like all air transport will move

electric here. And I think hopefully in our lifetime, it started out with like very short range flights, mostly because of like a specific battery energy density issue. We we have like if you had to compare like a battery versus like what's in an aircraft like, say, kerosene, there's basically roughly a 30 X difference like in like net energy in those two systems.

So you basically have an issue where like you can basically just go like 30 X like like like less far. And then with traditional with with with eVTOL aircraft, you're taking off vertically using a considerable amount of power doing that versus, like I said, an aircraft that might be going from San Francisco to L.A. that might be taking off conventionally. But we're starting with like, you know, we're starting now with like these short the kind of shorter flights helping them basically like move move passengers around.

It'll then happen regionally and then it'll happen at much larger, like longer, longer distances. The technologies needed to go like kind of like much longer distances are very different than what we're designing here today at Archer with eVTOL aircraft to get to like, you know, those level distances where you need to kind of like city to city traveling distances. You're going to have to redesign a different types of aircraft system to do that, in my mind.

Will that happen, though, do you think? It'll happen for sure. Yeah, for sure. It'll happen.

So if you could bet on a train, a car, a traditional plane or an EV flying vehicle going between San Francisco and L.A., you would bet on a flying electric vehicle? Electric, it's like, I mean, like we can design like some of the safest forms of transportation in the world, like flying in the air. I mean, flying today, we're like it's the safest form of transportation we take as humans. There will be like a large portion of folks that will travel between those cities like this in the air.

I think there will also be a large portion of that that also travels on the ground, whether it's train or tunnel or car. There's so many humans now and so many humans are just moving into cities. So you just have like a real issue where like the infrastructure is not just like able to support this at this moment.

Right. And how did you transition from Archer to Figurier? Could you tell us about that? Well, I think like one thing is interesting at Archer is like the aircraft is basically like a flying robot. And if you think about what we do at Figurier, build robots, you need, you know, you need batteries and motors and embedded systems like sensors, control software, a lot of the same kind of ingredients needed to kind of build like kind of any robot just in general.

So, you know, having built up Archer over the years, like got really familiar with those key kind of areas and integrating hardware and software together just overall. I mean, for me as an entrepreneur, like working in hardware, I think like the humanoid robotics are solving like general purpose humanoids is like the ultimate metaproblem. I think the end state for most things done in the physical world is done through a humanoid robot at this point.

I think we'll see billions of humanoids on the planet in our lifetime and they will do everything for us from things at home, like doing the laundry, washing dishes, unpacking groceries, cleaning up the house to we will see billions in the commercial workforce in areas of like logistics and manufacturing and health care. Our core thesis here is that the world was designed for humans, like we as humans designed it for ourselves. So everything we everything you do, we do all day is is designed for the way we look and feel.

And the whole world was for it. Even, you know, we're drinking coffees and teas, right, like with the cups that we hold and the door that we like exit from here and the stairs and all the tools we use are all designed for our, basically for the human form. So our belief here is that the humanoid is like the ultimate general purpose machine.

And so we have here is we have a deep focus on designing like really incredible hardware systems. At Figure here, we design basically the entire hardware and software stack on the robot, almost entirely ourselves at this point. So that goes down to like the motors and battery systems, all the electronics, wiring, kinematics and structures and sensors.

And then two is we we spend a large amount of time on the A.I. side. This is just like an incredibly hard problem. If you think about like a humanoid robot, the ones you see behind me, every robot has about 40 different joints roughly on the robot and a joint has like a motor that can spin roughly 360 degrees.

So the amount of the amount of positions the body can be in is equal to like 360 to the power of 40. And so that means that there's more potential states the robot can be in than atoms in the universe, which is like always crazy to think about, but it's like much more. So you can't solve this with code.

You have to use advanced A.I. like neural nets to be able to make this really work well. So we spend like a large fraction of our time outside of hardware on just like how do we get neural networks to control the robot and do the things we would need to do, like real stuff that humans would do. We're about three and a half years old now.

We've shipped now three generations of humanoid robots and we've been able to demonstrate that humanoids can do like useful human like work now fully autonomously and in most cases, like with neural nets all the way down the stack. I thought of this recently, I didn't realize how close we were to humanoids being in the actual world. And when I thought about that, I'm like, wow, I should have really looked this up earlier, researched it more because it feels very, I don't know, dystopian or utopian to picture a world where I have something that looks like me walking around my house, probably sitting in my office and in the car sitting beside me.

It's kind of a weird feeling to get your head around, maybe because I'm not used to it and I will get used to it soon. Yeah, I think like it just feels like such a sci-fi movie a bit, like when you see these

things working, it feels just so like it almost feels like it's not possible to go do and make it work. We have them walking around here in the office and it's just like still not like used to it.

Just like it looks incredible, like seeing robots out and about just like walking all day and doing stuff. Does it feel scary, Brett? Not to us, no. We feel really energized being around the robots and seeing it, not scary.

I mean, we built them, like we really understand the limitations and we've inherently tried to intrinsically design the hardware for safety and also the whole system for safety. So not scary to us. Like we're around them all day.

We've made them, we birthed them, like, you know, we brought them here. So it just like it's something really special about it where it doesn't feel like it's possible. It should be like 20 years out in the future.

But somehow it's like they're here and they're working really well. I think the robots we have are working really well. And to see that happen, you're like, oh, man, just give us five or 10 more years.

What would that really look like? It feels like it's just too soon for the world a bit. And but it's all happening now. Like we're really spend a lot of time trying to build like true general purpose, like be able to talk to a robot and have it do most useful things you'd want it to do.

That's like a really hard problem. Like it's like largely a software and AI problem at this point for us to go solve. But we can do it.

Like if you we like we announced, you know, recently, figure three, our third generation robotics right behind me. And we like we put out a bunch of content about how the robots doing a bunch of stuff in homes and even in like the more like workforce in the robot is doing all those tasks. Like it's doing all the manipulation for those tasks and actions all through a neural network.

And we did maybe, you know, a dozen dozen half or so use cases there and under in a few weeks with neural nets at places the robots never seen, you know, before a month before that. And I just like that's the ingredient you really need to scale at really like like large levels like you need like neural nets on humanoid hardware. And we're seeing that now.

So it just shows us how like how close we are. I don't it's not something we'll solve in a month, but it's certainly not something that's 10 years away at this point. Can we start with you breaking down a robot, Brett? What's inside? I think the first thing you could maybe start with is vision, because I did a little bit of research on vision and how a robot is seeing and how it has spatial intelligence in a way, but would love to hear from you.

Like I can see the robot behind you. What's in it? Yeah, let me start with like the basics of the robot and build up to this in terms of vision. Vision is kind of like higher level in the stack, but low

level on the robot.

It's basically a bunch of motors and we have like roughly 40 of them on the robot. We call them actuators, but they're like similar electric motors that we would see out in the world today. We have a bunch more sensors and transmission and other things in the in there that make it much more complicated.

But you can think of every joint as having an electric motor inside of it that is getting commands from a central computer and like what like how to output torque or how to basically move joint positions. That is powered by a battery system, like basically a traditional like lithium ion battery pack that you might see in an electric car or electric aircraft. And where does the battery sit? Yeah, it's right here in the middle of the torso on the chest.

So we have basically like a bunch of motors, we have a battery system in the torso and then we have a computer, both a CPU for compute and we have a GPU for inference for our AI models sitting in the torso next to the battery system. And how much of the compute happens physically on the robot and how much happens outside of the robot? Today, all of our compute happens all on the robot. So we do everything on board.

Will that change tomorrow? Some of the compute will start moving more and more off board. There is naturally some compute that needs to be done here on board. We need to run a decent amount of the neural nets at kind of like basically a couple hundred hertz, a couple hundred times a second.

There's no way you're going to be able to do that fully off board. The robot would just be too slow. You can generally off board.

You can maybe get a, you know, you could maybe get two to ten hertz or something like that. You know, you generally can communicate a few times a second, but we need like 200 times a second minimum on board the robot. So we have a lot of like on board compute today.

So it's on the robot to tell it how to move its motors and limbs really fast speeds. You've got to think like every little time, second, the robots is repositioning its entire body to balance its reposition, its hand and fingers, maybe heads moving to start seeing some better. And it's doing that closed loop, meaning like based on where the hands going and what the task is doing, the robots reevaluating what to go do.

And it's doing that, you know, a couple hundred times a second. So we have an actually a lot of compute here. We will off board.

We can off board more and more of the kind of bigger brain thinking stuff over time into the into the cloud. But today we do all on board, mostly because we don't need a we don't have to worry about having a network connection like you don't need Wi-Fi or 5G in this case. And we can

operate at really fast speeds and we know we can do the work like a human.

You know, I guess some humans can't can't work without a phone or network. But like, you know, humans really can do stuff without like a real Internet connection. And that's a similar how you would think about a robot to be a bummer.

The Wi-Fi went out and the robot can't do anything anymore. So our computer isn't on board. We have a ton of wiring, a ton of distributed electronics.

And then we have a bunch of sensors inside the motors and the rest of the robot to understand how to balance and feel forces and track positions. So like traditional robots that you see out in the world today, if you put your hand harder and harder on the table and keep pushing while not moving, the robots really can't really understand what's really happening. Our robot can feel that force like a human could.

We have we have basically torque tracking and force feedback in every joint across the whole robot. And that's really important to help balance and we have to touch the world and go out there and grab things. We really we need to understand those forces.

That's also a tricky sensor to integrate into the robot here internally. And yet we have cameras. So we see in pixel space on this current generation, we have, I think, eight cameras.

We have one on each palm of the of the of the hands. And then we have them in the head, the rest. So we can kind of see around ourselves.

So the robot sees through a camera. That camera can update, you know, like pretty quickly. And that feeds into an onboard computer that then ultimately tells what the motors what to do.

So that's, you know, roughly what's really happening. So the robot's reasoning in pixel space. It's it's going around like a self-driving car.

We don't we don't have LIDAR. So like a camera based self-driving car. And we're like we're reasoning through the world based on that to do work.

I'm not an expert at this, but if one were to say the Moore's law is starting to fail where the number of transistors you can put on a chipset will not increase at the pace that we have seen, they won't continue to double. Long term to increase compute on board. Is that possible to have enough compute on board to make the robot never need to connect to a network per se? Yeah, we run fully on board today in commercial settings every every day right now.

So without a network. So we have a commercial customer here in the US we've now been running with for over five months now. Every day we put this out publicly.

It's with BMW. That's a fully on board on board system that's doing all the reasoning on board of

what the robot should do. What you're seeing in the world is even if Moore's law slows down, compute is getting better.

So you'll be able to put more on compute over time than you have before. So our current figure three robot has more compute, like better compute than we had in previous generations. Second thing is like AI models that we're putting on are getting smarter and it's weird.

AI models are at one time getting bigger, but also some of the smaller models are getting smarter and better. So that's also in our advantage, meaning we can take kind of bigger like reasoning models that traditionally have been maybe a trillion parameters or whatever, billions of parameters. These can be quantized now into some smaller models that we can run on board that are also smarter than they were previously a year or two ago.

So we have models getting smaller and smarter and we have compute getting better. So those trends in general are in our favor. And today we're running robots fully on board, so we don't have to wait for like, oh, let's hope the compute catches up or models get some point in space.

We can do it all today. So we really today have a path where, you know, fully on board we can run robots daily now and in commercial operations. The big bottleneck for us is we're trying to solve general robotics here at Figure.

And that for us means like putting a robot into like a home it's never been in before and just do like a speech prompt, like a voice prompt, like ask it to do something. Like this could be like, I'm going to be gone all day, make sure my laundry is done before I get home. And you want that task done and done correctly over like a very long time horizon.

If I were to think every task, the ingredient or the input is energy. Like me as a human, I can eat a sandwich and I can do X amount because I can convert that into, I can convert the energy of that sandwich into something. Do you think the robot or the humanoid ever gets to that point where it becomes more efficient than a human in utilizing energy or creating energy? Today we're like far less efficient than a human.

Humans are like, we are like extremely efficient. Like our, you know, biological neural nets are really efficient. The way, how much energy or how much power we use to exert like walking and movements and things or, and even just sitting here idle is much better than we get on the robot today.

I see no reason over time, you know, we're seeing power drop now across our robot systems. Like, you know, there's a bunch of work that we're doing now. There's like, there's like the computational work to get more efficient.

And then there's like the mechanical work to get more efficient. Overall that we're looking at, both of those two were like, we're getting improvements on year to year, like our, our robot to robot. I

think we will approach, we see a path like approaching human, like kind of energy consumption.

It's going to be really hard. Like you're talking to us now in the early days of this, like we're like, you know, it's like way more like 16 years ago. It's like, it's, it's early.

It's just now happening. We, there's a lot of stuff on our roadmap that we're like, oh man, we got to go like solve, you know, manufacturing at scale, all these different things. And reducing power footprint is, is certainly one of them.

I wouldn't say it's a blocker today. Like we can, like we can do work in a lot of cases, almost the whole day, depending on what the task looks like. And if it's not that, if it's, you know, a task that has a lot of locomotion and movements, we can do kind of like human shifts in terms of time span.

Is that the ultimate test? Like if a robot were to have general intelligence, with the same amount of energy, it should be able to do not more of one task, but more of any task a human can do? Yeah. I think like ultimately, yeah. Like we would like maybe judge this on how much Watts it really took to do these, like, like, uh, like, like longer horizon tasks.

I think the near term thing is like, if you've got a robot in your home and you speak and you voice prompt it, like what percentage of those things can they do fully and then like a human and learn from those mistakes. Like we make a lot of, we do a lot of planning, like doing laundry. It's like, okay, well, uh, like where's the laundry baskets or what is the laundry other than the rooms? Like which rooms? Okay.

This room upstairs or downstairs. And you have to do a bunch of like, like really quick, like probably subconscious planning that's happening. And then we got to go out from like, actually, you know, uh, almost like serially, uh, complete those tasks.

There's a lot of like local failures that are happening. You go to a room and there's a laundry basket there. What do you do? Like a robot's got to be able to reason through that whole thing from A to Z. And then in some cases it could like for laundry, it could take hours.

Like the ultimate test right now is like, can we get a humanoid robot to do those types of things? And we've seen that in like LLMs. We've seen like a, like a, a robustness to like almost like human-like intelligence that we're like, we're tapping into. And we're trying to do that now in robotics.

Like how do we, how do we get this, like this machine to reason about the world? Like, like, uh, have like, um, common human common sense. And, um, I think that would just be like a, that'll be incredible. When we hit that, you're going to have a robot in your home and you're going to be doing this and it's just going to be incredible.

And then from there, yeah. Getting to more efficient systems. Um, getting the cost down.

I think you're going to spend a lot of time with the robots. Like it was like almost like a companion. Like they're going to be in your home.

They're going to be with your kids. They're going to be, they're going to know everything. Like they're going to have like almost like this infinite context window for memory.

Like, I think you're going to like talk to it and it's going to know if you like it's emotionally from an EQ perspective, it's going to know if you've had a tough day from just the sound of your voice and when you come home. And, uh, I think, um, there's a really important like program. Do you program EQ through, how do you teach a robot to have EQ? Does it measure your voice every day? And you have to like give a data as to this day I was feeling a certain way.

It learns like that. The best like AI, like, you know, um, calling native audio voice reasoning models today can do a decent job at like really understanding EQ. Well, I would say it's like, it hasn't done as well with like, I think IQ has been a big focus for like the whole space now of like information retrieval for voice models and things.

But, uh, there is a real path to getting voice models to like really high IQ, a high IQ. And, um, that can be a really understanding your tone and your emotional state when talking. Uh, these are like fully solvable things.

Uh, it's in the world model. It's in the datasets and it's a matter of like accurately training new like AI, like voice and reasoning models to be able to understand this well and do well at this. But this is a, this is a solvable thing.

We were spending time on here internally. I would say in the next year or two, you will have like voice, like AI models that you're talking to, whether it's on a phone or it can be through a robot or whatever, that you really won't be able to tell if it's a human or machine anymore. And going back to the robot, you said there are 40 joints, 40 motors, eight cameras, the torso has the GPU and the battery.

What else is in it? We have an order of like a thousand parts. So it's like, it's not just like, I, uh, as in Petrus is like, say, here are the things. What's in the head? The head has like cameras.

We have like the normal network connections you would want, like 5G Wi-Fi, Bluetooth. We have in figure three, we have three screens. We have a front screen, like an OLED, we have like two side screens.

We can like, I'll talk about that. We have a, we have a speaker and mic, a new speaker and mic system. That's like really, actually really good for figure three.

So we're spending a lot more time as like speaking to the robots. I think, um, like just voice is going to be the natural UI into the robot. Like you're not like, maybe you have an app on your

phone or whatever, but like voice is how you're going to communicate most of the time through the robot.

Um, and if you're not with it, you'll text it. Like the, like, like that is going to be such an important part of the ecosystem here. Is that the new form factor for AI overall to communicate with AI? Because a lot of people have been extrapolating what could be the next form factor after the cell phone.

Do you think it would be something voice related? How does a man communicate with a computer tomorrow? Yeah. My view is that like voice is the natural, like UI into artificial super intelligence here in general, whether it's like a digital systems or physical systems and voice is going to play an extremely important role in all of this. I think you will have both language devices, like almost like next generation, like kind of like iPhone level products.

And I think you will have humanoids and I think they will be networked really well together. Whenever I say this to my friends, they ask me if you're sitting in a meeting, you can't speak to interact with your device, but you can pull up a phone. Yeah.

The, the reality is like the, there's a couple of things like as we think about like the next frontier of devices and stuff, uh, the first is like speech is just not good enough today. Like our voice models are not good enough today for us to really understand how this could be like ubiquitous and it's like the main UI. They're just like, they're, they're B, they need to be a plus.

They're not good enough in a lot of areas. You, they talk over you, you can't interject the, sometimes the IQ gets collapsed in training and it's not very intelligent. And when tool calling, there's a bunch of run random issues, but that is going to get really good over the next 12 to 18 months.

What we will see in 12 to 18 months, voice models approaching human Turing tests. And that's not just from an IQ perspective and information retrieval. It's, it's EQ as well as we've just talked about naturalness.

Um, so I, I think, I think, um, that's one, two is like the, the models today just don't have enough context. We're missing like the, like when we're like going out in the world and we're talking today, like all this conduct is being missed. Like the, the, the models don't understand this.

What it's forcing us to do is it's making us go to like a computer and type it, type in prompts and ask it things and tell it like, Hey, I'm like, I have this thing and please help me here. But if you had like all this information, like you had like a limited memory, uh, and retrieval of all this information we're having all, all throughout the day, every day, uh, these AI systems would need like very little understanding of what, what is like currently in context or like what you need to do right now to help you. My view is like the next generation of stuff is basically language devices and humanoid robots.

And that, that is like, that'll go extremely far in my mind. Will they replace the traditional phone or computer? Yes. Every phone and computer was built pre AI.

The, the, all the hardware we have today in the world needs to be completely rebuilt from scratch. Think about it like this. We have like human-like intelligence in like the digital format now.

And we are communicating with this through like really old pre AI devices. There's zero chance or very low chance that there, they were somehow built specifically well enough for this new technology. In my mind, there's, it's not, they're not optimized well enough for this.

So you need a whole new class of devices. And robotics is a part of this story to basically be able to take this, like, what's happening right now is we have like this massive AI overhang where AI models are able to do incredible things. Uh, in like, uh, I would call it like the tech community.

Like we have like humanoid robots, and incredible things with AI. We have like, I can ask, you know, computer using agents to do crazy things now on computers without any human involvement. And then most of the world is using AI systems for information retrieval, like a Google search.

And it's mostly a constraint on hardware at this point. So this is a digression, Brett, but I was investing in a hardware company recently called Nothing, the phone company. And give me some advice.

I was playing off this thesis that a lot of the large language models are getting very democratized and the world will end up at a point where they're all a commodity and they're very similar to each other. And hardware will be the new differentiation. Does that kind of make sense? It's a huge differentiator today.

Cause like people are, like a lot of folks are just scared of building hardware. So you have like a lot less folks just even spending time on it. And then hardware is just like so much harder to go build than software.

So you have like a bunch of like this, like collapsing conversion funnel folks that like who are spending time on that really knows what they do. And then most people work on hardware, build crappy hardware. So you have like a problem where like there's just not a lot of good hardware in the world and a lot have failed trying to do this.

And, you know, at this, at some point you will buy a robot because of what it can do from a safety and costs and performance. Like you want a robot that can't just come in and just do one thing in your home. You want it to do everything and be fully general purpose.

And then you want to get even better in learning. That's a data, that's a data problem. So the more data we're feeding into the system, the smarter it's going to get, less mistakes it's going to

make.

You want like a, you don't want like the dumb robot in your home. You really want a smart robot in your home and everybody does, right? You don't want the dumb employee at work. You only want the like the fastest logistics robot or like the, you know, the highest performing manufacturing robot.

So at some point the mode becomes like, what can the robot capabilities look like? That's a data problem in some ways. And you need the hardware and AI systems to do that. You need the models and you, the AI in the, in the robots, but ultimately becomes data.

I think same with those devices you were talking through, like at some point, these models will know everything about you. It's almost like a really good executive assistant, a really good executive assistant, like just is really high performance. But like if you swap that person out and get a new person the next day, it's like, it's back to zero and it's really hard to get off the ground and start working.

So I think data is going to be incredibly important. Um, today we're not capturing most of that throughout the day. So there's a huge opportunity to really capture this in my mind.

If you do compute on board and you're competing with, say a large language model, which is doing compute in a data center, for example, or anywhere, will that integrated with a piece of hardware, maybe not humanoid, but any kind of hardware, not be smarter than a humanoid? The onboard model, you mean, versus like the off-board model? Yeah. We can run off-board models today. This is not a problem.

Like we, we've actually done off-board models. So we, we have, one thing we have in the robot, like on GPUs, we have it. We have a B, we have an LLM on the robot.

We have a vision language model, very similar to what we would like all use today, um, out and about in the world. We use that for like semantic reasoning. We use it, um, to really understand the world around us.

And, um, if we can find a better model off board, we will use that. Like there's no technology, um, blocker for us using like a trillion parameter model in the cloud, if that makes sense. But do you think that's the way forward? Is that where you would be five years from now, off-board? I think you will certainly have off-board reasoning.

I think you will have, I think you'll have both. I think you'll have off-board, like a big brain off-board that you need to talk to that can do very complex, um, maybe planning and reasoning. You will have onboard something that can move like much faster and much more reliable.

And then you'll even have smaller, even smaller models living on board. We call it the S1 level, like system one level that are doing like very fast, um, kind of like really quick closed loop motor

control. If I were to buy a robot and all of the compute and memories on board, isn't there a certain amount of learning beyond which it can't learn anymore because it's capped to capacity? There's an area of like fleet learning here that's really important.

We are taking data from all the robots and we are consulting that out to like our servers here at Figure and we're training on that. And what we're seeing now is we're seeing like incredible transfer from other robot data that's helping the current robot do better, which is like, which is unbelievable. Like moving packages on like a logistics line is somehow helping us fold laundry.

It's like the numbers get better. I watched that video by the way. It was really nice.

Very, very interesting. Yeah, thanks. I mean like that type of stuff is helping like other tasks.

And it's just like, that's kind of crazy, right? Like the only technology we've ever seen that work and it's like, is, is like, is deep learning and it's working. And we saw that with LLMs a bit. So I think that's one.

And two is you really want the robot as it's making mistakes in the real world, understanding and learning from that. Like having a reward for it. So like you really want the robots like, you know, screwing a cap on top of a bottle and it's messing up.

You really want it to understand how to get better at that over time. It's what humans do. You do trial and error, kind of like self play.

And so you have like an aspect of like fleet learning that you really want the robot to learn from. And two is, you want the robot to get better as it's doing tasks locally over time. The thing I keep taking away from more and more is the near term thing that we're seeing that's like been just absolutely astounding is we've been able to take one model, put all data into it and see all tasks get better.

And you know, we basically like really strong transfer. And that's that's a really important here for general purposeness. We really want the robots to do a vast amount of things and we want the, we want the fleet to continue to learn from it.

The good news is like, I don't know if you have kids, but I have like three kids. They all like, they all kind of need to like learn on their own, like learn to walk and like learn to do their own thing. You can't like tell them how to do it.

They have to kind of like learn through a trial and error. That's not how humanoids learn. Like once one robot knows how to do like logistics, that's the same neural networks we can put to every single robot in the fleet and they can all do it.

And so you have this weird scaling law here for humanoids where once like one robot knows how to do it, every robot in the fleet, if you have millions, millions of robots at that point would be able

to really understand how to do that task is going to make scaling just absolutely insane. Once we kind of figure out how to like really run this flywheel of data acquisition, flywheel and loop and training loop really quick. It's going to be able to like robots are going to wake up every day and know how to do new tasks in the world.

Could one argue that because you are generating the data to learn from, I would say it's synthetic data in a manner. We'll never compare with data in the text form or, you know, data I can scrape off the web. It has significantly more data points if I were to teach a model on language versus here the robot doing something and learning from it.

Are you trying to make the point that like learning from like texts, basically you're saying like text is so much more dense than like maybe like, yeah, you know, I think they all serve like a little bit different purposes. There's a lot more data out in the physical world than there is in the internet. We've kind of like we've like tapped out all the kind of tokens on the internet at this point.

So we think like being able to explore the physical world is like extremely important for like overall intelligence. There's like, you know, there's a thought process here we can go through of like if we're, if we're actually not able to like change our environment a bit as humans, like can you really achieve like real, real intelligence? And so we think, we think like touching the world and interacting with the world is like extremely important on the path like artificial super intelligence for humanity. And it's like a, it's a very large data set that we need to go out and collect.

I think the, as really said, like text pre-training that we've seen in the world that I think that's going to be extremely important for semantic reasoning and almost like this, like, like, like language, like language condition bridge into the human world. We're able now to talk to the robot and say, I was, I was in the lab last week. We're, we're, we're just like, you know, we're collecting data from humans navigating different types of apartments and homes.

We're taking that, we're training like a navigation model, basically a model that we can tell the robot walk to this place in this apartment. And it was a lab. And I basically asked the robot to walk to the oven and there was a table and stuff and chairs in the way.

And I just like, you know, told it, it walked all the way around the chairs, walked to the oven. It was perfect. And then I went, I went to the back and grabbed like a very large ladder.

It was like this big orange ladder. And I put it inside the apartment. And I positioned the robot away from it behind a table and asked the robot to navigate to the ladder.

And this ladder is like fully out of distribution. It's never seen a ladder before in its whole life. But like, you know, like a tech space LLM or VLM has seen a ladder.

You know, you could ask, like, you could ask like a, like a, like a, like a language bot, like, Hey, is this a ladder or not? They'd be like, that's a ladder. It's an orange ladder. So like the semantic

reasoning is there, but the robot has like the LLM has like no idea how to tell the body and joints how to move to approach the ladder and get there.

And, um, I watched the robot at that point, just like walk fully to the ladder, which was like really incredible. I think it was like a natural part of the, this, this, this part for us of going to solve general robotics that the semantic reasoning we're getting from like pre-trained language models is, is, is like, um, vital to basically be able to pull this off. And I think there's another piece, which is like how do we actually interact and touch the world, which I think is another piece of like a raw intelligence that's needed for humanity.

And I think the goal for us is to be able to kind of bridge those two worlds and basically build the right world model. So the robot can like navigate the world effectively in the physical world, which is like much more higher dimensional than the internet or text. And you spoke about kids, Brett, would you leave your kids with your humanoid? And also this whole companion concept, how important is it that the robot doesn't look menacing or does not look scary? Does the robot of tomorrow have, does your humanoid of tomorrow have facial expressions and stuff like that? Um, so actually I've had a robot in my house now for like a few months.

And, um, you know, my, my kids have been there, but we like, we monitor the robot. So we're not like, we're not like a, we're like, we have somebody with the robot. Like we're not letting the kids go up and hug it.

We're like, we're just doing tasks. We're trying to understand real robot performance through safety for me is like having a robot that we've designed in my home with my kids who have young kids and like fully autonomously. And we're not, we're not there today.

Like I would not let my robot roam free for hours and weeks right now with my young kids. There's a, there's a certain track record of safety and performance that we, like I would want to see internally before like doing that. I think that's just like a really high watermark for me, uh, of trying to get there.

And we're, we're pushing like every day to try to achieve this. Um, and then I think like you, you touched on a few things on the design side. It's funny, like it's almost like, does this like end at Westworld is what you're trying to ask.

Like maybe like, um, and it certainly seems possible to go build this. Like you've seen, like I've seen like, um, robots with like, you know, real like human-like faces and expressions and, um, like really good. I would say, um, our view right now is that we are trying to go out and do useful work.

We're not trying to fool another human that we're a human from like a entertainment perspective or maybe companion perspective. Uh, so we've not designed the robot with like facial expressions and like, uh, facial features and things like this. We've designed the robot to be like

basically a tool for humanity.

Um, there's another question is like, do we approach Westworld or not? Which I think is, um, I don't know. I'd like, I'm in some way I'm like really inspired and want to try to push there, but we're also like on a little different track here at figure today. I think we definitely don't want to like design a robot that is like a scary or aggressive or any way, but we also don't want to like try to like put like googly eyes on it and make it and fool you that it's like a kid toy and it's not.

And, um, if you go back and look at the early way most, it was like, it had the big googly eyes. And I, I just like, I, um, I quite despise that like design language. It's almost like, um, this is a very complex and sophisticated technology.

And you're trying to put like googly eyes on it to fool everybody that it's like, it's like a, it's a toy. So I think our approach is like. Have you seen those toys like Moxie and stuff like that? Which those are small.

Those are small things. I mean, our, our robots like, like, uh, over five foot, it's like, you know, over a hundred pounds. Like it's like, it's a, it's a real thing.

And I think if we were to start putting like googly eyes on it and high, like bright colors and stuff and make it feel like a toy, I think it would do a disservice to, uh, you know, our customers in the world. I think this is, we need to, we need to approach this like a, a very high end technology, uh, which it is. And, uh, we, we want to market it and we were designed for that.

We don't want to make it scary, but we don't also don't want to make it feel like it's a, you know, it's, it's not a kid's toy. So I think we're trying to figure out how to like, like basically marry that design language here internally. It's, it's tough.

We're getting better at it. As you're seeing the robots behind me, like it's radically changing. Like the robot over here is like made of all metal and silver and wires hanging out.

This is figure two here. Um, and then over here is figure three. So that, you know, we're seeing like radical advancements in the design, uh, like an industrial design for figure robots.

And I think, I think that will continue. I think you will see us approaching what we think is our ideal design state. Um, so we'll see.

We're also in the early endings of figuring out what user feedback looks like and where to go. So we don't have like all the answers for this. Uh, so it was like, you know, I guess a question for you.

Do we, do we approach Westworld or do we stay on the current track? What's your view? I think Westworld. Okay. Westworld.

Cool. Yeah. Cause why not? Right.

I think, why not? Yeah, let's do it. Yeah. Yeah.

Somebody will do it. Yeah. That'll for sure happen.

Um, it's possible, which I think that would be pretty crazy. We, um, I think if you put like a, I think even our robots now, if you put like a wig on it and put a jacket on it and it was like standing, like, you know, with the, with the back to you, I think you would sometimes maybe like say, is that a human or not? So, um, because it's walking around and doing, it's grabbing like, you know, water or coffee at the office or things like that. You'd be like, um, so I think there's no reason that we couldn't approach Westworld.

I'm sure there's no reason why we couldn't, my we couldn't put my face or your face on it either. At some point. We're already doing my voice.

Like I can like talk to the robot and say, go to the breath voice. And then it goes to the breath voice and talks to me with my voice. It's a, it's a crazy thought.

Imagine losing someone close to you, like say a parent or something like that. And creating a robot which looks just like him, speaks like him and has his memories, which are in this particular robot programmed in that, that seems like Westworld plus plus. It's just starting.

You know, we're like, we're like, we're a few years into it now. And I think over the next 10 or 20 years, it's just going to be, um, it's going to be a sci-fi future, which is going to be really fun. So Brett, I, I speak to a few AI people in the Valley, like a lot of the really popular ones.

Do you think a lot of them are speaking about dystopia or a dystopian world in a manner to create regulation so they have a regulatory moat in place so other people can't compete so easily? Do you think they're talking dystopian a lot right now? Is that your view? I would say it's divided half. I would say 60% dystopian, 40% utopian, but I sometimes question if the dystopian narrative is beneficial to them if a regulation were to come out and not have many others compete with the incumbent leaders. I think it's a little hard.

I think my like inner circle of folks that I spend time with in the office and things are so highly optimistic on the future of what this can do. Like too much pessimism is a little bit of a poison. So we have like an aggressive optimism here at the office and a value for it, the company.

Like my whole inner circle of like folks that I spend my time with, even outside of the company and stuff, are just so optimistic on like a positive future that we want to live in. I think it's the first step is just trying to think about like what is that positive future and trying to approach that. Like I don't have any interest in spending like most of my material life on a company that's going to ultimately help it be worse.

That wouldn't make any sense. So I think, I think this is all headed to a really good spot. I think there's obviously some chance like these things could go bad, which is always the case in all technology and things that we're doing in the world.

I think that probability is nonzero. I think we've got to be very, very thoughtful about making sure we don't approach that or if we are approaching it like safeguards in place. And also then painting the picture of what like a positive future for humanity will look like and approach that like, you know, as atomically approach that as fast as possible.

So for me, like, I know, humanoid robots for me, we look at this, we're like, this is going to be incredible. We don't want to do, you know, we want to like help the world in so many ways. And it's going to be like, who wants to do dishes? I, and do like, I woke up this morning and was like helping to put away dishes in the dish.

But I was like, cause my kids were eating like breakfast. Like I don't want to do that anymore. It's terrible.

And, um, I don't think anybody really wants to do that kind of stuff. So I think that stuff is going to be really critical. I think the AI systems that we're seeing today and like advanced reasoning models and are going to be like extremely beneficial for the world, uh, net, net, net positive.

And, um, I don't know, I'm rooting for that, for that future. So I'm heading down like that track, but I certainly understand, uh, there's a, there's a good amount of folks sitting, sitting there. Just like I would say scared and maybe rooting against that.

Brett, you didn't cover eyesight. I, I, I tried to read up about eyesight and I watched a bunch of people talk about it, but I couldn't understand it. How does eyesight work? Yeah.

So we basically, it's super simple. Our input to the model is like the, the robot eyesight, um, what, what the user is prompting or speaking. And then the joint states like where the joints are, like where the robot's at and in time.

So like, if you take like a sliver of timestamp, you'll get a, you'll get a, you'll get an image which is coming from a video frame. You'll get the, the prompts, like what's the prompts, like go do dishes or whatever, walk to the ladder in this case, because I said before, uh, and then what's this current state of the robot? Like that's the input state. And then the output is basically like, uh, the action space of the robot is, uh, sending out a targeted commands, uh, for, for in joints.

Like where, where do I position my body to go do? And that's running like really, that loop is running really fast. So the robot just like literally sees what the humans see like from the head. For, for me to understand in a simpler manner, you're not saying it's like a way more, which is more radar and LIDAR, but like a Tesla self-driving car, which has more cameras.

Yeah, exactly. So we almost like, if you think, if you put like a VR headset on and you can like

see the world through a headset, that's what the robot sees. And like very, very similar.

We have like two cameras on the face and they just like point out and they can see it. We, um, in our launch video for figure three, we have like when we started doing the laundry, when it grabs one of the pods, it's actually the video coming from the robot head. And it's just like, it looks like you put a camera, like a GoPro on somebody's head and they're like, they're moving around the world.

That's what the robot sees. And so it just sees like, you know, kind of like sees a egocentric video, uh, similar. If you had a GoPro in your head, it's hands, it has a prompt of what to do as a goal in mind, like a behavior.

And then it knows its state and then it sees what it needs to do. So in a lot of ways, it's like visual servoing at scale. If it needs to grab the cup, it sees a cup, it needs to grab the cup as a command.

It knows the state where my hand positions at head, torso, body. And then it sees its hand as it relates to the, to the cup. And it just seems like, you know, it needs to approach this.

And it's almost like building like a closed loop planner on how to go do that. And that's what the neural net's doing. It's outputting commands of where to position the body to go achieve this target.

And it's doing this over like, you know, thousands and thousands of demonstrations of humans of, you know, doing this, doing this exact work or similar work. And then we're, we're using that for training. So, um, our training set is like human data.

So we use humans for navigation and manipulation. We're like, um, you know, humans are wearing, uh, we have humans like basically doing this work. We're recording this information and we're training Helix, which is our neural network internally to be able to do this on the robot.

What has changed, Brett? Like maybe a year or so ago, I had Yann LeCun, who's one of, you know, he worked on the initial AI projects and is quite a authority on it. He came and explained to me how a transformer was created, uh, how the, how the generative AI models came about. I'm quite an idiot and I couldn't understand all of it, but what has changed in the world that is allowing for a humanoid to be built today that could not have been done five years ago? Well, one is like deep learning just works.

And I think that's an extremely important part of this. We are able to take very complex things like a complex robot with high degrees of freedom. We talked about earlier.

We will output actions out of it using deep learning models. That's the, that's the thing that like didn't exist five years ago to really make happen. Didn't exist the right way for us to make happen.

Uh, but also like the hardware, like we have like really good inference compute and training compute now to really make this really happen. People don't talk about this a lot, but the hardware has really gotten good. Um, if you go back and like watch, like old, you know, humanoids like 10 years ago, a lot of them were hydraulic systems.

They weren't even electric. They were like leaking oils. They're extremely dangerous on humans.

Uh, they're very big and bulky, sometimes loud. Yeah. Yeah.

I was, I was watching this Boston, uh, Dynamics founder speak about how he still loves hydraulics because of how much power output he's able to get out. Yes. Yes.

Like you weren't able to get enough power out of these systems 10 years ago. Uh, the power density, torque density wasn't really high enough. So you had to choose some other technology, which was hydraulic system at the time.

We've now had torque density and power density of actuators and battery systems get good enough where you can run like fully electric systems instead of hydraulic systems. And that makes the robot smaller, lighter, way quieter, faster. Uh, like you can run the run times longer.

It's just like a better, it's like the, it's where you want to go. When you go into market, you don't want to bring to market a hydraulic system. There's also like local sensors on the robot cameras, torques unseen, other things that like we've spent a lot of time on.

Like they were kind of, some of these were like do or die products internally last three years that we've advanced internally ourselves that we think are probably state of the art. Like nobody else in the world has done systems like this that has allowed us to, to run neural nets in the backend. So I think like, I think hardware is one.

So like electric hardware, I would say specifically. And then I would say the deep learning and algorithms and compute around there have been fundamental breakthroughs that weren't possible 10 years ago. In a way, would you also say the big change has been a humanoid can today leverage prior knowledge in a manner that it could not? A hundred percent.

Like it has semantic grounding. Uh, it has memory. Uh, it has like a language condition.

So humans can tell it like to do something, like grab the coffee cup on the table, put it down. It knows exactly what that really means. Like from a human, doesn't need like a translator to the robot world.

It automatically understands that knows it's the coffee cup that I talked about from, from like a language, like a language prompt or voice prompt. So, um, all of that, you switch it all together. Like, wow, it looks like it's possible now to make this happen.

It looks like general robotics is like actually possible today. You know, not yet working 10 hours a day in my home, but it does work in my home in some cases for like these like individual stuff. That's hardest.

Like we can do laundry. Like if you think about the things you would need to see to make general purpose robots work, the hardest thing in that book is doing like high rate manipulation of deformable objects, like, like laundry towels, like things like this, like folding a towel, for instance, that like constantly moves, you grab it. It's not like, um, it's not rigid.

It's just like a really hard problem to solve. Or it could be like an unlimited amount of variations a towel can be in. We have neural networks that can do that now.

And that for me felt like the last chapter of the book. So we can fold towels that neural nanohumanoid that we have solved in robotics. The screwed up thing is that we've actually already solved.

We already know how to do that. We can fold laundry. And there's other things in the ecosystem that we're working on now to get that working fully end to end.

So it's like it can work for hours like that without any human intervention. Some of these chapters now the book are filling out, some are left open now and we're just like, we're trying to fill the whole whole book out in some way. But like the stuff that you would ask me three years ago, if we were doing this podcast when we first started, I would have said like here's a little laundry I need to go fold and we can do that.

I'd be like, we've solved general robotics, like we're there. So in some aspect, uh, we've like solved some hard parts, but we still have a lot, like a lot more to go. But we, we see it now.

We see like a little light in the tunnel to go do this. And what comes out of that tunnel is like a robot I can put in your home that can do 10 hours or 20 hours of work a day and do the things you don't want to do all day long. What's happening with the others in your industry, Brad? Like I've looked at physical intelligence, they seem more like they're building the software more than the hardware, Neo Gamma, Boston Dynamics, Tesla.

Who's doing what, which is very cool. I'm going to play the video of your robot doing the segregation of the packages, but what are the others doing, which is quite cool? I don't know. It's kind of been interesting.

I feel like some of the older generation of robotics companies in the space, the last like three years have in some way kind of collapsed. Groups I've like, you know, 10 years ago I would watch and look up to, I think it was just like, I've kind of like somewhat faded away. I think it's been this emergence of like new, new companies now in the space that are like, I think that are quite interesting overall.

The thing that's been most interesting for me is like, everybody's trying to make some sort of bet. And there's a lot of like, you know, in like China, you're seeing like a lot of just hardware only bets. Like they're just building hardware.

They're selling it. They're generally not working on hands. They're generally not working on AI systems that much, but they're not, there's no like full stack solution.

It's like, here's like, you know, you can go to Unitree and buy a robot hardware today. It's like, go, go buy it. And then it's up to you to go make it work.

It's like there's not much you can really do with it today. And then on the opposite side, there's folks that are just working on pure AI, like no hardware. And you mentioned some of these here that like, don't care about the hardware, care about the AI systems.

In our mind, you really can't choose one or the other. You have to just like, you have to go down a road where you, you do the hardware, you do it incredibly well. You do the AI, you also do it incredibly well.

And then you put it all together really well. Like the companies who can do really well are going to do both really well. And what they're going to output is a robot that can do useful work like autonomously.

Like what you want to see is a demonstration of a robot doing something that's like not tele-opt or like coded. And what you're seeing now in the world is like, I would say most major competitor to us at this point are putting out most of their content and updates tele-opt from a human. And I think it's like perhaps some of the most deceiving things I've ever seen.

It's like if I was driving a self-driving car, if a self-driving car pulled up next to us and we found out there was some guy from Tennessee driving it, we'd both be like, that's a, that's not what I thought this would look like. Can I push you a bit on Elon? The way to review Tesla's stock as a multiple of earning to the price the stock trades at, suddenly the humanoid is starting to form a part of how one would evaluate the stock. What are they trying to build? I don't know what they're trying to build.

They're building a humanoid. I don't know anything more than you know what's public. Like internally here, we kind of like put like a little bit of horse blinders on in terms of like our focus to what we're doing.

You know, we're kind of reasoning from the ground up of like what we should really do. We don't really pay too much attention to like what the competitors are doing. I think that would have been, I think that would have ended in a pretty badly last three years if we did.

I think like for instance, like a lot of groups in the space don't work on like high dimensional hands

and we made a big bet internally, like we're going to go all in on hands, make them over time. They'll be complex and they'll be extremely important for high, high dexterous manipulation. Tesla should be really great at this.

Like, so I'm rooting for them to do well. What do you suspect will be the first problem a humanoid figure three will solve or figure four or five or whatever? First real world problem. Since April we've had figure two in our first commercial customer working every day, every work day at the client for 10 hours a day.

It's doing a real task like on the real production line, helping to help build cars. This is not like a high volume. So we're like, we're, but the point is like we're learning how to operationalize robots.

We're learning how to get robots to run every single day. How does it work autonomously? Is there any human interventions? Is there hardware failures? Like are there software failures? Like how do we make it better going forward? And we've gotten now like close to six months of like real world data. We think maybe the one of the first in the world to have done this of a robot running every single day in the commercial workforce and like really tough environments.

It's not like it's easy. That's been good. I think for figure three we're making a much bigger push now into the home.

Than ever before. So we think the home, I thought you were focusing on factories and auto plants and stuff like that. And companies like Neo Gamma, we're focusing more on the home.

We've made a big, big push starting last six months into the home here. And we're making an even larger push on figure three into the home. Figure three will happen first home.

Because the factory will not probably need the humanoid form factor because the factory can be rebuilt or remodeled to support. I don't believe that thesis. No, like you go into the factories, they don't do, you could rebuild a factory from scratch, Greenfield and then you'd still have tough time removing all the humans.

It's like you see, you see some of the most automated manufacturing and logistics companies in the world. And they're like, they're heavy human. They're heavy human.

I think here to build like millions of robots in the world, our view is that the humanoid is the right path to go make that really happen. Listen, I think three years ago you would run this call three years ago. I would have said like the, the home is such a harder problem in the commercial workforce because it's much more variable.

Like both of our homes will be different. They're going to another home. It'll be different, like different toasters, different appliances, everything.

So you need like real common sense reasoning in the home that you might not necessarily need

to ship into like say a manufacturing logistics company. They won. We've like, so our view is like shipping in the commercial workforce buyer, what buyer time to solve the home are my view.

And this is completely flipped last year. I think the home is near term. I think a home is solvable.

I think we're basically data bound this point. So we launched a program two weeks ago called project go big. And we're trying to go build the largest kind of like internet scale, pre-training set for robotics in the world.

We're trying to build basic data to teach the robot how to do things across various tasks in the home. Every week now we're collecting data in different new homes and apartment buildings here in the U S and we're using that for training Helix. I think this next 12 months, I think we'll probably build the biggest robot retraining set in the world to go make this happen.

It's like LLMs in some way, we're just, we're just data bound. And then the more data we're seeing, the better the models are getting. So in our, in our side, we really can't scrape the internet for robot data.

Like LLMs can, we have to go generate it. And the good news is a humanoid is like, like a human. So we can like capture this from human data.

We've already been in my home last like two to three months in like alpha testing. We've, we've, we've showed some of the results last week of like what robots can do in homes with like, you know, in those cases, those tasks were all done through Helix. You know, we will do more of that over the next 12 months and we will also spend time shipping robots commercially.

We think it's nice to get outdoors and get robots out to the real world. It's really messy and learn how the robots will behave. It's in a huge, huge benefit for us.

Whenever I speak to people in the US I've been spending a fair amount of time there this year. Everybody talks about China is so far ahead in robotics. Why is that? Is that manufacturing at scale? What are they doing that others can pick up on? Yeah.

I just don't believe this. There's a lot of chatter about this constantly. Like I look at like kind of manufacturing 2.0 and robotics 2.0. I'm interested in like the new generation of manufacturing, the new generation of robotics going out.

And when you start looking at that space, the US and particularly figure is ahead of every single Chinese company I've, I've, I've seen globally of what I know publicly in terms of capabilities. Like watch the robot do advanced things with neural nets autonomously over long horizons. And the answer is like, there's really nobody else in the world even close to what we're doing there.

Um, there is a ton of manufacturing capacity in China, but the problems right now are not solving manufacturing in this problem. Solve the problem right now is solving a general purpose

humanoid robot. Once you solve that, then it becomes a data and manufacturing constraint.

But we are in a race to solve general purpose robots. We are not in a race against China and manufacturing robots right now. And it doesn't matter.

China makes a million humanoid robots tomorrow. They just don't work very well to be frank. And maybe they will in the future.

But what I've seen today is they're, they're very far from even where we're at at figure. And I know the work they need to go do to get to where we're at. And I know the work we need to go do to do general purpose work.

It's not going to happen tomorrow. It's not going to happen next month. There's a, there's a large like 14 peaks we have to go climb to go get there.

So then once we transitioned from there, we need to go have to make, make a lot of robots and get good at manufacturing. What the manufacturing looks like here, it looks like consumer electronics manufacturing. And I think when people think about manufacturing and get scared of it, they get scared at automotive manufacturing.

Automotive manufacturing is like the most complex systems I've ever seen in my entire life, all put together to manufacture cars. It's like hundreds of times harder than manufacturing consumer electronics. Like we manufacture consumer electronics a lot by hand in the world and we ship billion phones, whatever it is.

And it's making cars is extremely tough. Making humanoids is not a car. It's a consumer electronics.

You can hold every partner hand. You could theoretically build millions of these by hand. Uh, we won't, but you can.

So the constraint for me is not like if we were sitting here today and trying to invent the next like AI device or next phone, like small phone, you, we wouldn't say like, well, the hard part is going to be making a lot of them. We would say getting the AI systems and the hardware designed and, you know, engineering and prototype is probably the harder parts making them is like, you know, we don't sit here and say like you're going to die making like consumer electronics products. Like everybody's gonna be able to make it.

I think humor is a similar way. I think you can manufacture these. We're doing them now.

I think, I think manufacturing is hard, but less hard than automotive. And I do not think it's whoever has the most manufacturing capacity wins this game right now. I think it's whoever has the best general purposeness.

And that's just that the hardware design layer and the AI layer and the overall integration of those systems that are useful work. So right now America is winning this. I think globally our goal is to beat China at this.

We're working seven days a week here to go do that. And like, you know, at the end of the day, I'm like pro human. So I hope like wherever happens, like we we go out and like build incredible things for humanity and make it really work.

But I think it's been a mischaracterization that like, oh, China's got this capacity. Oh, China's got this headstart and it's hard to keep up. That's completely wrong in my mind.

I read somewhere that you decided to move away from the incumbent hyperscalers, the models, like you had a partnership with OpenAI which stopped in 2025. Why is that? The short answer is we'd like we OpenAI led my series B. We generally were working on some kind of advanced models with them. For maybe a year or two years, a year or so, but a year and figure ultimately chose to kind of leave them and go in our own direction.

The summary is that we were just way better at this stuff internally. We were good at getting models designed and embedded on the system and running on a robot and working. We were just really good at it.

And I think we've been a good at it and we viewed the right path was just be able to do it ourselves. Most of the advancements we made during that period of time were all done internally. And we just felt the better path for us internally would be to design a system, our AI systems ourselves, which we've been doing basically since the start.

It also became difficult for us. Like even, you know, we have a really large AI team here and they're like some of the best AI robotics folks in the world. It also became hard, like hiring people when we had a partnership like that, where folks felt like we were outsourcing our AI, which we weren't doing.

And we were actually doing all of it mostly ourselves internally. And just like for many reasons, it became like, this is not the right path for us to go down. The path for figure is to go alone and do an extremely good job of bringing it here internally and understanding the models really well down to the substrate levels, and then be able to put it on embedded systems and make it really work.

And it's not just like sitting in a room like training models and like throwing it over the, over a fence and making it really work. It's, it's, you gotta really, it's, this is hardware. You gotta be able to sit there with the robot and get it to work.

And that, that is just like a, a, it's a lot of trial and error. It's a lot of evaluations. Um, it's a lot of hard work, but at the end of the day, the buck stops with being on robot and making that stuff

work.

And the team here is just the best in the world to be able to do this. They, I mean, they're from, from DeepMind and like some of the best places in AI and they're, they're sitting here with the robots. They're sitting here with a GPU cluster and training models.

And we're doing that daily now and evaluating what it looks well, like really good. I sit here and watch us do generally anywhere between like five and 20 evaluations of new models every single day. And like, we've been doing that now for a year and a half, two years, like every day we're testing new things and trying to kill climb that that's kind of led to the results you've seen today.

And then what you haven't seen is like the next generation of results that we're working on in here internally. And um, I just can't wait to show those as well. Brett said, my primary job is that of a investor, stock market investor, public market investor.

When I look at the world of AI, if I were to extrapolate and say world output is about a hundred trillion and let's say half of that comes from services of one kind or another. So say 50 trillion in services for the amount of money that has gone into AI, we haven't seen the revenue from AI commensurate with that. When does that change and how does that trickle down into humanoids? Will it be the same or similar for many years? I mean, my view is this will take some time, but like AI systems will just eat everything.

So software ate the world prior to this and AI will eat software in the rest of the world. And it'll, it'll take some time. It's hard to kind of take these existing habits and things that are embedded in society and radically transform them over like a very short period of time.

But over like five, 10, 15, 20 years, I think things will radically transform. My view is that there's going to be a generational company to be built that will be able to put ultimately billions of robots on the planet that will build like the largest company we've ever seen in the world by a long shot. And it's driven from like the metrics you said, you know, I think it's a little under half of GDP is like human labor.

It's not just like getting to the point where like, okay, you have the biggest market in the world now, which is like human labor. And you're like, you can basically build a human, a mechanical human. How many like cheap humans that work 20 hours a day would you want if they're like extremely affordable and they can do human-like performance? And they're like, you know, one 10th the cost of a human or whatever it would be over time.

Like a limited, right? As I keep putting more humans, like more mechanical humans in, and they keep giving me ROI output. You'll just keep feeding them into the economy. And GDP in a lot of ways is like a kind of a, this by-product of like human productivity.

And we even measure GDP on a per capita, like per human perspective is how we measure

GDP. So if the denominator there per capita per human is like per synthetic human or per humanoid, I think you have like this unbounded effect to GDP that it's not just like, okay, the biggest team in the world, but that, that TAM could like grow, like by many factors here in the future. So I think it will take time because we can't just like snap our fingers and like appear, we just talked about manufacturing, but we can't snap our fingers and make a hundred million robots.

But I think through our lifetime, we will be able to build billions of robots and get them out to the world. And they will ultimately start building their own robots from themselves. What will emerge on the humanoid side will be the largest company in the world by far.

And then on the AI side, you sense, I think AI is just going to eat everything. And advice to an investor who is looking at the various AI companies, let's say open AI, Anthropic, Alphabet, all of these big incumbent players in the space. If you had $1 to invest, where would you put it and why? Is it private or public or both? Either.

Either. Personally, I put all my money in my companies. Like I don't make public investments.

I put it all in here. So I would tell you biasly, put them into my companies, but listen, I'm a big believer in kind of like growth, investing, investing things. I really think I believe in, I'd be at the frontier for me of deep tech.

I think it's. And which company is that? It's just like deep tech would be like, I work in deep tech. I mean, Archer and cover and figure my companies that I've started are deep tech companies.

And they're like at the frontier of like really hard things look possible. And if solved, I think it'd be like radically important for a lifetime. But could you pick a company which isn't one of yours? Like from the ecosystem? No, like the, the new age LLMs, I think those will go really far.

The open AI is an anthropics and I love, you know what SpaceX is doing and Tesla. I think these groups, I think these are like, you know, I think I, you know, I'm in the Bay area. So Waymo is a good example.

Like wait, the way most are everywhere here. It's like one of the best products I've ever had. I weigh mode be on my list.

Like these, these will be like generational companies. Uh, they're certainly on the track. To be like generational industries and then finding the right players inside of those that are like willing to play, you know, play hard for a long period of time and win.

I think these are like, uh, areas that I would be going all in on. I'm spending my life going all in on some of these. And I think that's like, I always like think it's important to be excited about what you're like, say, investing or spend your time in.

Um, so I'm a more believer of like kind of those areas. Anything else? Don't want to pick one. I

mean, listen, I would probably pick Waymo as my, as my pick.

I think I, you know, I think it's like a 50 billion market cap, uh, super underrated. Like, I think if you haven't experienced it in a way, Mo, you wouldn't really quite understand. I have, I have.

I've used it quite a bit in the valley. Yeah. What was your experience? Incredible.

This one particular time I was at a traffic signal in a way more and there was a fire truck coming behind me and my way more actually broke the signal, did this maneuver, went to the right and allowed the truck to pass. It was incredible that a self-driving car could do that. The conditional wisdom says this is like not possible.

If you go out and like talk to a lot of people, it's like, you know, self-driving is not quite here. Then you're like, right away. And you're like, oh man, it's been here.

It's here right now. And it's unbelievable. And it's going to keep getting better and better.

And I think of that for my kids too. My kids are young. I don't, I don't want them to drive.

Driving is too dangerous. I have some close friends that are senior Waymo. Like, I think Waymo would be great.

I think the market cap is reasonable. I think they've really proven their operational readiness to be able to make it work. I think there's a long ways to go now to scale that out to like a global fleet of like millions of cars.

I think the recipe is really there of like how to go do it. And the learnings are definitely there. I think the stack will evolve on the sensors and models and stuff over time to be more scalable.

And they've, you know, I think they've been at it for like 16 years now. So it's like, it's like almost like where this come from. They've, there's been people that are working for 16 years in this problem.

And that's also like, I think really inspiring to see folks like really playing for the long haul. Playing their own game and delivering a product that's like super complex. Like they have the complexity we have, but they don't, but then they have the safety thing of being able to not hit the road unless you know for sure you can make the right decisions at like fast speeds.

And you know, when we're walking at a max a meter a second, we don't really have those same safety conditions that a car would have in those cases. So it makes it even harder in terms of complexity that they have than we would have here. So being able to see that level of complexity, I think it's just astounding.

So Brett, my community is largely Indian origin entrepreneurs across the world who are looking to build something. If I were to extrapolate that humanoids have the biggest use case in societies

with say a declining demographic dividend or really high labor cost. For countries like India, where the cost of labor is still quite low compared to say America, for example.

And if I were to be a bit more candid, the access to risk capital is significantly lower here. What do we build? What does my community build in the humanoid space? What I can relate to on my side is like I grew up on a farm, came from very humble beginnings, had a no to negative money most of my life. And I think a couple pieces of advice I have just for folks in general is like I have a big thesis on this, and I was like just going for it.

I spent a decent amount of my career as an entrepreneur like de-risking into areas where I felt like are possible, especially in software that weren't like the really big, big ideas. And I think with Archer and Cover and Figure now and stuff I'm working on, these are big ideas and I could really get to go for it. I've learned a lot from that experience.

I've learned a few things. One is I think when you're going for these big ideas, they generally have bigger markets, like bigger TAMs. And I think investors like to fund bigger, harder things that have bigger outcomes.

They just like that binary risk-reward. So I think you just access the capital. I've seen it's generally easier when you're working on harder things that can have bigger outcomes than working on smaller things like a calendar app or whatever it would look like.

I think the second thing is people from an employee base that want to work for you, you need to hire really good people. They want to work for the really hard problems. They also don't want to work for a calendar app.

They want to work for a hard, really important problem for the world. And then I think the motivation and dopamine levels of folks that work on these things are also harder. And it's generally not like 10 or 50 times harder.

It's generally like, we're working on stuff that's probably like multiple times harder, but the reward if it pays off in terms of like, is like tens of folds higher. So one thing I would say is like, just try to go for it. And I know that also depends on access to capital and the cost of that and talent and everything else.

I understand that. But like, I have like, you know, my life, I just like, just try to go for it. I think that's one.

Two is the humanoid space, like, it needs enormous help. We need help on compute and training. We need help on data collection significantly over time.

We need help on components. Like, one thing I've really realized is three years ago, I thought there was a more of a mature supply chain. We can go out and get motors and cameras and

sensors.

Like, the world has tons of tens of thousands of different types of motors. Like, and what we found is we've had like, none of them are very good when you bring them in. And you start looking at it, like, the best motor we had, our actuator was like, twice as big as we needed it to be and twice as heavy.

We're like, ah, man, what are we going to do? We talked to the vendor, they're like, this is what we got. If you want to redesign, it takes like two years and maybe we'll get it, maybe we won't. And I was like, oh, man, we got to go do this ourselves.

The huge area of supply chain that needs help on. We need help on go-to-market activities as well over time. You know, I think what's been maybe more challenging with humanoids is that we've had to vertically integrate almost everything now, which also makes it tough for a community of folks trying to help out.

Like, you can't build on top of the humanoid similar to what you can build on top of with LLMs today and use them like an API. So we are literally designing the full stack. They can go into your home and do everything.

Down to like the AI models and the middleware operating system, the firmware, the embedded software, all the hardware. So it makes it a little bit more challenging for a startup to pop up and be like, yep, we'll use you. But there are opportunities today for this stuff.

How do we find these? I think one thing I have, you just got to talk to the right folks in the industry and then you got to start building stuff. You just got to go for it. One thing I always see for entrepreneurs is they have a hard time.

There's a lot of stiction, a lot of inertia to get started. Just go build some stuff. Go out there with a little budget, go build a little website, go out there and cold call your way through it.

You got to get out there and start trying some stuff. I think this trial and error period is extremely important. It's like a shark swimming.

If you stop, you die as an entrepreneur. So you got to keep just going. And you'll need to go do some trial and error.

You're going to talk to everybody in the space. There's a bunch of conferences for humanoids and embodied AI. You got to get into that network and understand their pain points, understand where the business is headed, what things those companies might need.

And then you got to start just making inroads into them. And you got to start building and showing value. For us, we have over a billion dollars of cash.

If there are things and problems in our landscape, we will go fund it and we will bring people in and go solve those. So if somebody approached us, hey, Brett or figure or whatever, we've done these things. Here's how we can really help.

We trial and test those frequently and see how that could work. I think it's just a lot of hustle. And if you're out there and you're listening to this, just go.

Just start going, start building. You can do a lot on a small budget and make up some traction. And that's how this all starts.

And then you learn a lot from there. And then you ask a bunch of questions, you get feedback, and you keep going and make that recursively better. I'm going to digress and ask you a question.

I've been exploring the idea of building a social media app out of India because I think the incumbent players in social media have gotten boring, for the lack of a better word. Any advice on what I could build around social media? I feel like I still don't have a platform that I really love, even here. I feel like there's certain aspects I can go to certain places.

I can go to LinkedIn and do some stuff professionally. You've got Instagram, you've got X. I think they all have very different experiences and ways to deliver content. And I feel like the world really hasn't figured out what the right platform is there yet.

I think there's something to probably do with an AI-native social networking platform that could be, I think, really important. And I don't quite live in that world to understand that well enough, but I think I would start with an AI-first native experience. And I think there are some gaps in what certain platforms are able to deliver today and also what the kind of content I want to consume and give out that I think can give rise to a new experience for folks.

I think we also need it. We're like, I'm burnt out, I don't use Facebook. It's just like, don't use these things.

We need something new and fresh and there's a way to do that. There's a way to get the multi-modality of maybe what I get across some of these other platforms into one experience. Any advice on who I should hire for something like this, Brett? I've always, I think the conventional wisdom of everything is to go out and hire somebody really experienced that makes you feel really good from a really good background that's at a successful company.

And I found that playbook is just complete crap. Throw that right out the door. And even now, Figures has gotten to a point where we have these big shots knocking on our door wanting to come work here now and big companies whenever else.

And it's just not the right, if you look at every generational company, it's not like they went out and

picked out the VP here and the VP here and the VP here and put them together. It's the opposite of what Meta, and Meta's doing that right now, right? Look at their Meta superintelligence lab. It's like you're putting 15 Tom Brady's together and making that work.

It's just going to immediately collapse. It's not going to work. And I think my view is you just need to find the core axiom of what I look for here for talent is people that really care.

And I can't find any better way to describe it than caring. They care a lot about whatever, you know, you're trying to hire somebody to do this. They obviously have the technical background to go do it.

But even if you don't quite have the perfect technical background, if you have a high aperture for learning, you're going to go out and figure out how to do that. You're going to learn to code or learn to vibe code or whatever you need to go do. So I have a big thesis on trying to find folks early stage that really care.

And that, for me, means a lot of shot on goals. It means getting myself out there, talking about what I'm doing and trying to find this big net, this wide net of folks that want to come in here and work with me on this really important project or whatever it is. And I think for you here, it's like finding folks that would really care about this problem.

I'm trying to build this next generation social networking app in India that's like native AI first. And I think the first step is like casting a wide net, telling everybody you're doing it. I don't think there's really any secrets in tech.

I think people hold those close to your chest, or they used to. I used to. And this doesn't matter anymore.

Like a lot of it's on just raw execution and ability to deliver product. That really works well. And most people can't execute and most people can't deliver a good product.

So there really are no secrets. So my view is to try to find somebody that deeply cares. Do you think it could be voice first? I'm like extremely bullish on voice models.

I'm doing work on some of this work right now. And I think for both like the digital AI front, it's funny, like we're doing like the physical AI thing, I figure. We're like, we're trying to do everything a human can in the physical world.

And like, you know, I've also been working on something that is also helping out the digital side of this. And all roads lead to full autonomy of AI agents, whether it's like through a robot or through like digital systems. And the UI to that, in my mind, is voice.

Like really good voice. And we just don't have like really good voice today, but it's getting better. Like the line's like getting better over time.

And it's like the gap's like narrowing of where we need to go. I think we're like 12 months away from like really good voice that can make an experience that you're talking about work. But we need it.

And if not, you don't have like a high tolerance for like a voice that you can't interrupt or makes mistakes. Like you just like, it's immediately a bad experience. But I think for your thing, we're like, I think where it's a 2026 event for you of getting voices, voice like voice models, like native AI, like probably half duplex models, like extremely good.

I think there could be a very significant voice aspect of this. And I also think about voice and generative AI, like rendering as a real differentiator to what's really happening today. And like kind of like 1.0 social media networks.

I would be starting to play around with a couple of different of these avenues and starting to prototype them within a week. I would want to see if you can use an off the shelf TTS for voice, start hooking up like a mock network together and start basically parallel pathing a couple of these like user journeys of what this could look like. And I would say it'd be nice if you could check in with me in a week and we could like start demoing these and checking and testing them out.

The only way through this for you is like to be able to rapidly prototype a very different type of ideas and feel the experience and what that looks like and trying to close the gap on what you think that experience should look like. Yeah, I'm going to hit you up after this and seek some help on this. I want to be your first user.

So I get like a really great user. And I can always say I was the first user on your platform. I'd love that.

Last question for you, Brett. Beyond this utopian future, when the humanoids are doing most of what humans can do today from a work lens or from even a home lens, what happens to the world? What happens to socialism? What happens to capitalism? What happens to labor? What happens to a job and getting a salary? I think there's a few things that happen. One is we're building autonomous, I feel like autonomous agents, like synthetic humans, whether digitally or physically, they will do everything behind a computer a human can.

And they will also do everything in the physical world through a humanoid that you would want to go do. If that's true, which I think is very high likelihood in the future, the prices of goods and services will slowly start collapsing, basically asatomically approaching zero. And you'll just have like an abundance of goods and services.

Like if you're like a robot farmer, all it would need is the price of land and then energy. And if the energy generated on site with solar and stationary battery, then you basically have like the rental cost of the robot and land. And it'd just be like, that would be the only real cost.

And I think the robot costs will like really dramatically reduce. You'll have this dramatic reduction here over the next like five years as we approach higher volume manufacturing. And then you'll have another step down once robots can fully build robots in the loop together.

And you remove all human labor from the whole manufacturing process. So I think you have that period of time that happens. I think you'll be able to afford anything.

Anything in the world that humans can do today, you'll have synthetic agents that will be able to do that for you. What happens then? I think there's a part of society that lacks purpose for what they do. Like myself, like what do I do? If like synthetic human can do everything I can do, just as good.

I spent my whole life career doing stuff that I think is valuable, that like a computer can do just as well. And then there's another aspect of us like freeing us up from all the stuff we do all day. Like we have, you know, we work 40, 50, 60 years of our life.

We work at home all day long doing laundry and dish. I feel like I'm cleaning up the kids' toys like completely constantly. Like just like, you know, just like, I feel like I'm working.

Well, I want to spend time with my kids, but I'm like, kids are getting breakfast and I'm like getting them breakfast or getting stuff out of the dishwasher. I don't want to be doing that. I don't want to be talking to them.

You know, I'm driving them to school. Like I just like, I can't, I have to focus on the street. You know what I mean? There's just like all this stuff we're doing that we're not really realizing is just like, and then I'm holding all these things in my head and planning like how to book something or go travel or send a wire.

Like I just want systems to go out and do this for us. I think it's quite freeing. I think when one aspect we lose purpose, another aspect is like extremely freeing.

Another aspect, it's like, you'll be able to afford anything in the world. And the funny thing is like, the weird thing is we're going to see that this in our lifetime now. I grew up on a farm, right? Like we're seeing things advance.

Like, you know, my grandfather didn't grow up around electricity in the early years. Like, it's just like, we're seeing things advance radically. I think it's going to be a net net gray.

It's just going to be a radically different 20 years from now. One could argue that humans didn't have purpose to begin with. We just deceived ourselves into thinking we had purpose.

100%. I mean, is it purpose like working like 18 hours a day every single day? You know what I mean? These are the things I don't know. I mean, it's really, I happen to like love what I do because I feel like I have purpose and I hope it's purposeful at some point in life, like over time.

Do you love what you do or do you like the byproduct of what you do? I love the byproduct of what I do, which helps me get dopamine and a feeling for like what I do every day in the office. So I really love being able to ship great products out. Nothing makes me happier than to see a robot working for five months out in the real world.

That's awesome. It's like, but I need, I want millions of that. But my days are hard.

My days, like all the, I have like this problem funnel. Like every problem funnel is right to me and I'm working on problems all day. I have this like joke where I'm telling people when I get to meet them, like tell me good news.

I've had a bad day. And it's just like, they never have good news because like things that are going well, I don't spend time on. But I'm working on problem fixing things.

I'm doing things at a CEO level that are needed to block the critical path for the company. So I just work on problems all day. I work like sunup to sundown.

I'm generally in the office every night. So like, no, my days are very hard, exhausting. I'm tired.

And no, they're like, they're extremely difficult and not fun on the surface. But like the byproduct of this is we get to create general purpose humanoid robots here. Is that dopamine from a place of ego where nobody's building humanoids yet and you're the first people doing it.

So you get a dopamine hit. Tomorrow, if everybody were building a robot, it might not be the same. Yeah, I think it really sucked not winning.

We want to go build something great. We want to go win. And I think it's really important.

Like we don't want to, I don't want to lose. Like losing is like, is terrible. Like we don't want to make bad decisions.

Like there's all this like, there's all this like thing in the Valley, Silicon Valley about failing, it's okay. It's like, it's not okay. I don't want to fail.

I don't like that way of thinking. Like we fail all the time and we make mistakes all the time, but we don't want to make those mistakes. I'd rather not have failed and succeeded.

Like, so I happen to make better decisions sometimes after I fail, but like, that's not the goal. The goal is to go win. I would say short-term, yeah, it's great.

It's great to go win. But like, even now, like this is a 10, 20, 30 year journey. There is like the winner will emerge 10 years from now, five years from now.

You know what I mean? Like not now. So like, whether we're like one or five or two right now, none of this matters. So like, what matters is shipping a product at scale that can generate the

data to increase intelligence and reduce costs.

That's who wins in my mind. Like who's the first to a million robots in market? That's a big marker that will help determine like the leader. So I think that for me, like, even though I think we've been successful today, we do not feel in any way like we're winning or I've won here.

I think we want to go really win it. And that's going to take five, 10 more years of beyond of more extreme like seven, eight week work that we're doing. So, but we're in it.

We're in it for the long haul. We started this company knowing it's going to be a multi-decade thing. And we're just fortunate to be here.

And I think we're most fortunate, I'm most fortunate as a founder being here. You know, like my advice to entrepreneurs is always like, always be within the right decade. And it was unclear when I started this company, I was like, I'm going to go do this for like 20, 30 years beyond if like humanoids were going to happen this fast.

And it happens to be that it's happening this fast now. And it's a race now to go solve general purpose robots. So I feel fortunate to be in a place where we have over a billion in cash.

We have great commercial customers. Robots are working with neural nets. Figure threes, I think it's the best humanoid in the world.

It's great. We have a great team. Team is like by far and away the best asset we have at the whole company.

So we have all the ingredients now to like pull off what we started as a dream. And I think we feel fortunate about that. Thank you, Brett.

Thank you for doing this. This was fun. I'll be down in the valley in November and I'd love to come check them out in person.

Come by and like give me a part of the network, social network here early too. Yeah, a hundred percent. We'll connect after this.

Thank you, Brett. Cheers. All right, bye.

Bye.